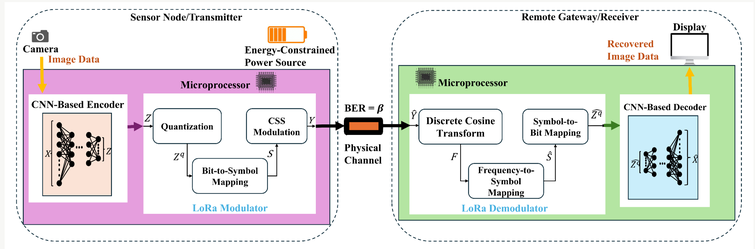
A Lossy Compression Technique Using Disjoint Autoencoders for LoRa Communication

Santhosh Manoharan, Kang Gao, Subir Biswas
NeEWS Laboratory, Dept. of ECE, Michigan State University

Introduction

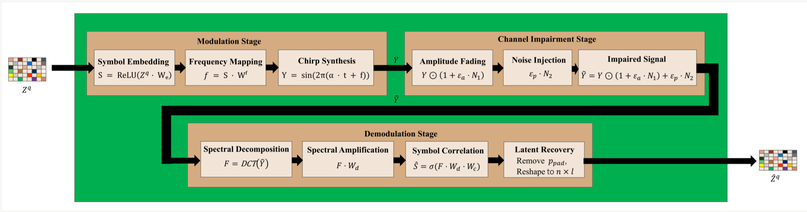
- LoRa offers **long-range communication on a battery**, but only **22 bps – 27 kbps** with payloads as small as **51 bytes**.
- **Images matter at the edge:** remote IoT nodes in agriculture, environmental monitoring and field surveillance increasingly need to send **pictures**, not just sensor readings.
- Classical codecs (JPEG, PCA) **break under bit errors**; deep JSCC is **too heavy** for edge nodes and trained on idealized channels.
- **Our goal:** compress images to fit LoRa's tiny payloads, survive radio errors, and still run on a **Raspberry Pi Zero**.

System Model



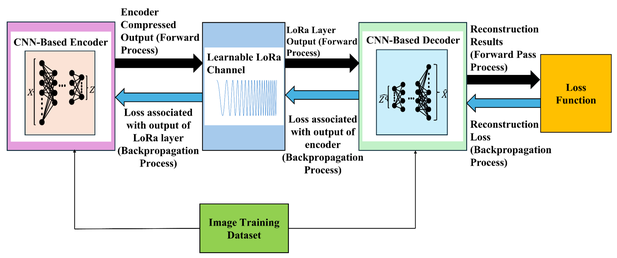
Disjoint encoder (sensor node) and decoder (gateway) over a LoRa link

- **Disjoint autoencoder:** trained jointly, deployed separately; latent size **n × l** sets bandwidth vs. quality.
- **Learnable LoRa channel:** differentiable chirp modulation, fading/noise, and DCT demodulation, used only in training.



Learnable LoRa channel module (training only)

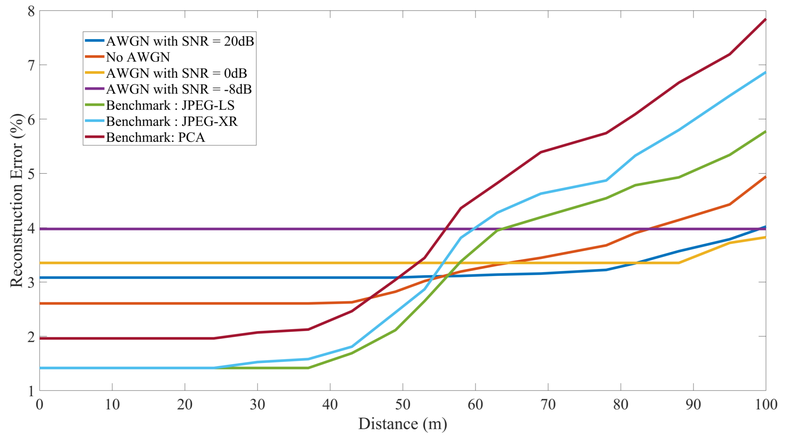
Two-Stage Training



Training loop with the learnable LoRa channel (offline)

- **Two-stage training:** 50-epoch pre-train without the channel, then 30-epoch fine-tune with the learnable channel at SNR 20, 0, −8 dB.

We trained image compression **to handle radio static**, so tiny sensors can send **clear pictures from far away**, where regular images **fall apart.**



Reconstruction error vs. distance on a live LoRa link (n = 32, l = 24). Legend SNR = training SNR of the learnable channel.

Stays clear at 100 m — Error holds at about 4% at every distance tested.

Halves classical error — 3.98% vs. 7.85% (PCA) at 100 m.

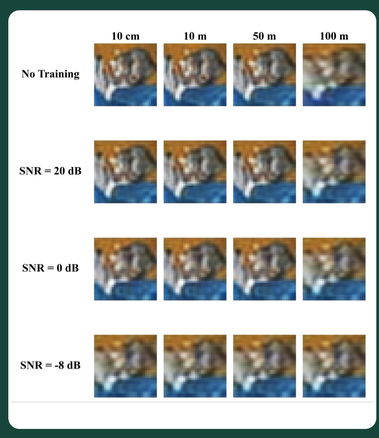
4× faster delivery — 15.5 s per image instead of 64.5 s.

Distance picks the model

Lowest-error choice at each range, read from the graph above.

0 m – 25 m	25 m – 50 m	50 m – 75 m	75 m – 100 m
Classical / plain AE	SNR 20 dB	SNR 0 dB	SNR −8 dB model: ≈ 3.98% at every distance (worst-case guarantee)

Seeing the difference



Received images at 10 cm, 10 m, 50 m, and 100 m.

No training: colour shift and softening by 50 m.

SNR = 20 dB: simulates ideal conditions, very similar to no training.

SNR = 0 dB: trained for harsher conditions; slight initial drop in quality, but more persistent with distance.

SNR = -8 dB: looks almost identical at every distance.

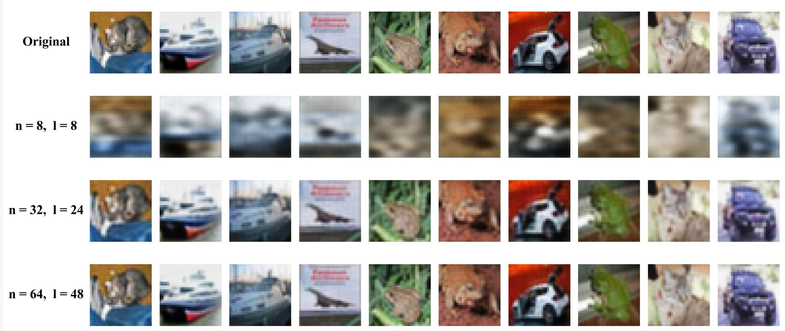
Scan the QR code to get the paper, slides & contact info

Santhosh Manoharan · manoha10@msu.edu

Why channel training helps

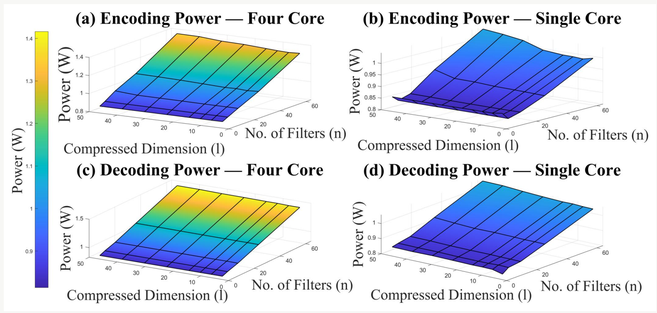
- During training, the model sees the same errors a real LoRa link produces, so it learns to work around them.
- JPEG-XR / JPEG-LS lead at 1.42% on a clean channel until 24 m.
- **At 100 m:** JPEG-XR 6.87%, JPEG-LS 5.78%, PCA 7.85%, AE w/o channel training 4.94%.
- **Channel-trained models at 100 m:** 20 dB → 4.02%, 0 dB → 3.82%, −8 dB → 3.98% (flat).

Quality vs. compression



- Error **2.01–3.14%** with channel training (n ≥ 32, l ≥ 24); robustness costs only **≈ 0.3%**.
- Bandwidth reduction up to **99.48%** (0.31 s per image).
- Larger n and l improve quality but send more data and raise computation cost.

Edge cost (Pi Zero 2W)



Encode / decode power vs. latent size, four-core vs. single-core

- FLOPs **0.5 M → 100 M**; quadratic in n, linear in l.
- Quad-core is fastest and uses least energy (**33.7–303.7 J** vs. 39–526 J single-core).
- Single-core only makes sense when other tasks need the remaining cores.

Experimental Setup

- 2 × Raspberry Pi Zero 2W + LoRa Bonnet, 7.8 cm monopole antenna
- 915 MHz, BW 500 kHz, SF 7, 23 dBm; 250-byte packets
- **CRC & FEC disabled**
- CIFAR-10 dataset; outdoor line-of-sight
- **91 distances** from 10 cm to 100 m, **250 images** at each

Future work

- Domain-specific IoT imagery
- Non-LoS / multipath channel module
- Runtime model switching from measured SNR